

## 1. Introduction to Routing

**Routing** is the process by which a router decides the best path to forward a packet from a source to a destination across a network. Every router maintains a **routing table** that stores this path information.

Routing algorithms are broadly classified as:

| Type                                      | Scope                                  | Examples                                 |
|-------------------------------------------|----------------------------------------|------------------------------------------|
| Intra-domain (Interior Gateway Protocols) | Within a single Autonomous System (AS) | RIP (Distance Vector), OSPF (Link State) |
| Inter-domain (Exterior Gateway Protocols) | Between different Autonomous Systems   | BGP                                      |

A network is also classified by routing approach:

- **Static Routing** – routes manually configured by an administrator; doesn't adapt to topology changes.
- **Dynamic Routing** – routes computed automatically using algorithms that adapt to network changes (this unit's focus).

## 2. Intra-Domain Routing

Intra-domain routing finds the best path for packets **within one Autonomous System (AS)** — a network or group of networks under a single administrative control (e.g., a college campus network, an ISP's internal network).

Two major classical approaches:

### 2.1 Distance Vector Routing (DVR)

**Core Idea:** "Routing by rumor." Each router knows only the distance (cost/number of hops) to each destination and the *direction* (next-hop neighbor) to send packets — it does **not** know the full topology.

#### Key Principles:

- Based on the **Bellman-Ford algorithm**.
- Each router maintains a vector (table) of: {Destination, Distance, Next Hop}.
- Each router periodically broadcasts its **entire routing table** to its directly connected neighbors only.
- On receiving a neighbor's table, a router updates its own table if a shorter path is discovered.

### Bellman-Ford Update Rule:

$$D_x(y) = \min \text{ over all neighbors } v \{ c(x,v) + D_v(y) \}$$

Where:

- $D_x(y)$  = estimated distance from router x to destination y
- $c(x,v)$  = cost of link from x to neighbor v
- $D_v(y)$  = neighbor v's distance estimate to y

### Working Steps:

1. Initially, each router knows only the distance to its directly connected neighbors (distance = link cost).
2. All other distances are set to infinity.
3. Each router exchanges its table with neighbors periodically (e.g., every 30 sec in RIP).
4. Each router updates its table using the Bellman-Ford rule.
5. Process repeats until tables converge (no further changes) — this is called **convergence**.

### Example:

A --1-- B --1-- C

- Initially: A knows  $\text{dist}(B)=1$ ,  $\text{dist}(C)=\infty$
- After B shares its table: A learns  $\text{dist}(C) = 1(A-B) + 1(B-C) = 2$ , next hop = B

### Advantages:

- Simple to implement and understand.
- Low computational overhead per router.

### Disadvantages / Problems:

- **Slow convergence** — takes time to stabilize after a topology change.
- **Count-to-Infinity Problem** — when a link fails, routers may keep incrementing distance values back and forth, slowly counting up toward infinity instead of detecting the failure quickly.
- **Routing loops** can form temporarily.

### Solutions to Count-to-Infinity:

| Technique            | Description                                                                                |
|----------------------|--------------------------------------------------------------------------------------------|
| <b>Split Horizon</b> | A router does not advertise a route back to the neighbor from which it learned that route. |

| Technique                                | Description                                                                                                                                   |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Split Horizon with Poison Reverse</b> | Router advertises the route back to the source neighbor, but with a distance of infinity, explicitly signaling "don't use me for this route." |
| <b>Hold-down Timers</b>                  | After a route is marked unreachable, the router ignores any updates about that route for a fixed time, allowing bad info to flush out.        |
| <b>Triggered Updates</b>                 | Updates sent immediately upon a topology change, rather than waiting for the next periodic update.                                            |

**Real-world protocol: RIP (Routing Information Protocol)** — uses hop count as metric, maximum of 15 hops (16 = unreachable), updates every 30 seconds.

## 2.2 Link State Routing (LSR)

**Core Idea:** "Every router builds a complete map of the network" — unlike DVR, each router has full topology knowledge and independently computes the shortest path.

### Key Principles:

- Based on **Dijkstra's Shortest Path Algorithm**.
- Each router creates a **Link State Packet (LSP)** containing: its identity, list of directly connected neighbors, and the cost (link state) to each.
- LSPs are **flooded** to *all* routers in the network (not just neighbors), so every router eventually has identical, complete topology information.
- Each router independently runs Dijkstra's algorithm on this topology map to compute shortest paths to all destinations.

### Steps of Link State Routing:

1. **Discover neighbors** and learn their network addresses (via "Hello" packets).
2. **Measure the cost/delay** to each neighbor.
3. **Construct a Link State Packet (LSP)** containing this neighbor + cost information.
4. **Flood (distribute)** the LSP to all other routers in the AS using **reliable flooding**.
5. **Compute shortest paths** using Dijkstra's algorithm once the complete map (link state database) is built.

### Dijkstra's Algorithm (brief):

- Maintain a set of "permanent" (finalized) nodes and "tentative" nodes.
- Start from the source node with distance 0.
- Repeatedly pick the tentative node with the smallest distance, mark it permanent, and relax (update) the distances of its neighbors.

- Repeat until all nodes are permanent.

#### Advantages:

- **Fast convergence.**
- No count-to-infinity problem (each router has full topology, so loops are far less likely).
- Scales better for large networks.

#### Disadvantages:

- Requires more memory (stores entire topology) and more processing power (runs Dijkstra's algorithm).
- Flooding causes more control traffic, especially in large networks.

**Real-world protocol: OSPF (Open Shortest Path First)** — widely used interior gateway protocol; supports hierarchical design using "areas," uses cost as metric (often based on bandwidth).

### 2.3 Distance Vector vs Link State — Comparison

| Feature            | Distance Vector                         | Link State                                             |
|--------------------|-----------------------------------------|--------------------------------------------------------|
| Algorithm Used     | Bellman-Ford                            | Dijkstra's                                             |
| Information Known  | Only distance & next hop to destination | Full network topology                                  |
| Information Shared | Entire table, only to neighbors         | Link state of itself, to <i>all</i> routers (flooding) |
| Convergence Speed  | Slow                                    | Fast                                                   |
| Problems           | Count-to-infinity, routing loops        | High memory & CPU usage                                |
| Example Protocol   | RIP                                     | OSPF                                                   |
| Scalability        | Poor for large networks                 | Good for large networks                                |

## 3. Inter-Domain Routing

When packets must travel **between different Autonomous Systems** (e.g., from your ISP to a server hosted by another ISP), intra-domain protocols like RIP/OSPF are not used. Instead, **inter-domain routing** governs this using the concept of Autonomous Systems and policy-based routing.

### 3.1 Autonomous System (AS)

An **Autonomous System** is a collection of routers and networks under a **single administrative authority**, using a common routing policy, and a single intra-domain routing protocol internally.

- Each AS is identified globally by a unique **AS Number (ASN)**, assigned by IANA/regional registries.
- Examples: an ISP's network, a large university network, a corporate WAN.

#### Types of Autonomous Systems:

| Type          | Description                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| Stub AS       | Has only one connection to another AS (e.g., a small organization's network); traffic source or destination is within it. |
| Multihomed AS | Connects to more than one other AS, but does <b>not</b> carry transit traffic for others.                                 |
| Transit AS    | Connects to multiple ASes and <b>does carry traffic</b> through itself between other ASes (e.g., large ISP backbones).    |

### 3.2 BGP — Border Gateway Protocol (Basic Idea)

**BGP** is the standard **inter-domain (exterior gateway) routing protocol** used to exchange routing information between different Autonomous Systems on the Internet. It is often called the "protocol that runs the Internet."

#### Key Characteristics:

- BGP is a **Path Vector Protocol** — it doesn't just track distance/cost like DVR; it tracks the **entire path (sequence of AS numbers)** a route has traversed, helping to avoid loops.
- Runs over **TCP** (port 179), ensuring reliable delivery of routing updates.
- BGP is **policy-based**, not purely metric-based — ASes can apply business/administrative policies (e.g., preferring routes through a paid transit provider over a free peer) rather than simply choosing the "shortest" path.

#### Two Types of BGP Sessions:

| Type                | Description                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------|
| eBGP (External BGP) | Used between routers in <i>different</i> Autonomous Systems.                                        |
| iBGP (Internal BGP) | Used between routers <i>within the same</i> AS, to distribute externally learned routes internally. |

### How BGP Works (basic idea):

1. BGP routers (called **BGP speakers**) form **peering/neighbor relationships** with routers in other ASes.
2. They exchange **reachability information** — i.e., which network prefixes (IP address ranges) are reachable through them, along with the **AS-PATH** (list of ASes the route passes through).
3. Each AS applies its own **import/export policies** to decide which routes to accept, prefer, and advertise further.
4. Loop prevention: if a router sees its own AS number already in the AS-PATH of a received route, it rejects that route (this prevents routing loops between ASes).

### Why not use Distance Vector or Link State directly for inter-domain routing?

- **Scale:** The Internet has hundreds of thousands of routes; full topology flooding (Link State) is infeasible.
- **Policy, not just shortest path:** Organizations need control over routing decisions based on business agreements (peering/transit contracts), not just minimum hop count/cost.
- **Trust:** Different ASes are independently administered; BGP allows policy autonomy rather than blindly trusting neighbor-supplied metrics.

## 4. Routing Tables

A **routing table** is a data structure stored in a router (or host) that contains information used to determine the best path for forwarding packets toward their destination.

### 4.1 Typical Fields in a Routing Table

| Field                      | Description                                                                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Destination Network</b> | The destination IP address/network (often with subnet mask or prefix length).                                                                         |
| <b>Next Hop</b>            | The IP address of the next router to which the packet should be forwarded.                                                                            |
| <b>Interface</b>           | The outgoing physical/logical interface on the router.                                                                                                |
| <b>Metric/Cost</b>         | A value representing the "cost" of the path (hop count, bandwidth-based cost, delay, etc.) — used to choose the best among multiple available routes. |
| <b>Flags</b>               | Indicates route type, e.g., U (up), G (gateway), H (host route).                                                                                      |

### 4.2 Construction and Updating

- **Static routes:** Manually entered by the network administrator; remain fixed until manually changed.
- **Dynamic routes:** Automatically learned and continuously updated by routing protocols (RIP, OSPF, BGP) as the network topology changes.
- **Directly connected routes:** Automatically added for networks directly attached to the router's interfaces.

#### Routing Table Construction Process:

1. Directly connected networks are added automatically.
2. Static routes (if configured) are added.
3. Dynamic routing protocol(s) running on the router exchange information with neighbors/peers.
4. The router computes the best route to each destination (using DVR/Bellman-Ford, LSR/Dijkstra, or BGP path selection) and updates the table.
5. The table is continuously refreshed as updates arrive (periodic or triggered).

#### 4.3 Metrics and Path Selection

When multiple routes exist to the same destination, the router selects the **best route** based on a **metric** — a numerical value representing the cost of a path. Common metrics include:

| Metric                  | Used By        | Description                                                                               |
|-------------------------|----------------|-------------------------------------------------------------------------------------------|
| Hop Count               | RIP            | Number of routers a packet must pass through. Lower is better.                            |
| Bandwidth/Cost          | OSPF           | Inversely related to link bandwidth — higher bandwidth links get a lower cost.            |
| Delay                   | Some protocols | Time taken for a packet to traverse the path.                                             |
| AS-PATH length & Policy | BGP            | Number of ASes in the path, plus administrative preferences (Local Preference, MED, etc.) |

**General Path Selection Rule:** Among all available paths to a destination, the routing algorithm selects the path with the **lowest metric** value (or, in BGP's case, the path that best satisfies configured policy first, with AS-PATH length and other attributes as tie-breakers).

#### 6. Possible Exam Questions (for practice)

1. Explain the working of Distance Vector Routing with the Bellman-Ford equation. What is the count-to-infinity problem and how is it solved?
2. Differentiate between Distance Vector and Link State routing algorithms. (Common 5/10-mark question)

3. Describe the steps involved in Link State Routing. Why is Dijkstra's algorithm used?
4. What is an Autonomous System? Explain its types with examples.
5. Explain the basic working principle of BGP. Why is it called a path-vector protocol?
6. What is a routing table? Explain its fields and how it is constructed/updated.
7. Compare RIP and OSPF.
8. Short notes: eBGP vs iBGP; Split Horizon; Hold-down timers.